Assignment 1

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Introduction

National Supported Work (NSW) Demonstration was a subsidized job training program spanning 1974 to 1979 designed to help disadvantaged workers including long-term AFDC (welfare) recipients, ex-addicts, ex-offenders, and young school dropouts secure stable employment (Manpower Demonstration Research Corporation, 1983). Historically, econometric evaluations of such programs faced immense confounding and consequently produced unreliable causal inferences. This paper attempts to estimate the effect of program enrollment on real trainee earnings where participants were randomly assigned to treatment (job training) and control groups (no training). By applying the Causal Roadmap to the classic LaLonde dataset (1986), we can follow the Causal Roadmap to define our assumptions and compare three statistical estimators: G-computation (standardization), Augmented Inverse Probability of Treatment Weighting (AIPTW), and Targeted Maximum Likelihood Estimation (TMLE) with Ensemble SuperLearner (SL). More information on the LaLonde dataset can be found in the documentation for the R package 'designmatch'.

Research Question

We are interested in knowing to what extent participation in the NSW Demonstration program affects future income among Black participants. In causal phrasing: if, perhaps contrary to reality, all eligible Black individuals were assigned to the NSW Demonstration training program, what would their expected real earnings in 1978 be compared to if, perhaps contrary to reality, none of them were assigned to the program?

Data

To answer this question, we observe the data structure $O = (W, A, Y)$ for n independent and identically distributed individuals.

- W is a set of 6 baseline covariates that could confound the effect of A on Y: age, years of education, marital status, absence of a high school degree, and real earnings in 1974 and 1975.
- $A \in \{0, 1\}$ is the binary intervention. $A = 1$ if the individual was assigned the job training, and $A = 0$ otherwise.
- Y is the continuous outcome of interest: real earnings in 1978.

Our sample is comprised of 243 Black individuals, corresponding to 156 treated and 87 control subjects, and 10 the aforementioned variables.

```
data(lalonde)

lalonde <- subset(lalonde, black == 1) # Only Black people for the ATE

summary(lalonde) # Everything looks good, no missingness
```

```
##    treatment          age          education         black      hispanic
##  Min.   :0.000   Min.   :16.00   Min.   : 1.00   Min.   :1   Min.   :0
##  1st Qu.:0.000   1st Qu.:19.00   1st Qu.: 9.00   1st Qu.:1   1st Qu.:0
##  Median :1.000   Median :24.00   Median :11.00   Median :1   Median :0
##  Mean   :0.642   Mean   :26.01   Mean   :10.23   Mean   :1   Mean   :0
##  3rd Qu.:1.000   3rd Qu.:29.00   3rd Qu.:12.00   3rd Qu.:1   3rd Qu.:0
##  Max.   :1.000   Max.   :55.00   Max.   :17.00   Max.   :1   Max.   :0
##     married          nodegree           re74            re75
##  Min.   :0.0000   Min.   :0.0000   Min.   :    0   Min.   :    0
##  1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:    0   1st Qu.:    0
##  Median :0.0000   Median :1.0000   Median :    0   Median :    0
##  Mean   :0.2222   Mean   :0.6955   Mean   : 2499   Mean   : 1614
##  3rd Qu.:0.0000   3rd Qu.:1.0000   3rd Qu.: 2048   3rd Qu.: 2039
##  Max.   :1.0000   Max.   :1.0000   Max.   :35040   Max.   :25142
##       re78
##  Min.   :    0
##  1st Qu.:    0
##  Median : 2821
##  Mean   : 5677
##  3rd Qu.: 8984
##  Max.   :60308
```

```r
lalonde <- lalonde |>
  select(-hispanic, -black)

summary(lalonde) # Everything looks good, no missingness
```

```
##    treatment          age          education        married
##  Min.   :0.000   Min.   :16.00   Min.   : 1.00   Min.   :0.0000
##  1st Qu.:0.000   1st Qu.:19.00   1st Qu.: 9.00   1st Qu.:0.0000
##  Median :1.000   Median :24.00   Median :11.00   Median :0.0000
##  Mean   :0.642   Mean   :26.01   Mean   :10.23   Mean   :0.2222
##  3rd Qu.:1.000   3rd Qu.:29.00   3rd Qu.:12.00   3rd Qu.:0.0000
##  Max.   :1.000   Max.   :55.00   Max.   :17.00   Max.   :1.0000
##     nodegree           re74            re75            re78
##  Min.   :0.0000   Min.   :    0   Min.   :    0   Min.   :    0
##  1st Qu.:0.0000   1st Qu.:    0   1st Qu.:    0   1st Qu.:    0
##  Median :1.0000   Median :    0   Median :    0   Median : 2821
##  Mean   :0.6955   Mean   : 2499   Mean   : 1614   Mean   : 5677
##  3rd Qu.:1.0000   3rd Qu.: 2048   3rd Qu.: 2039   3rd Qu.: 8984
##  Max.   :1.0000   Max.   :35040   Max.   :25142   Max.   :60308
```

```r
# Define Covariates (W), Treatment (A), and Outcome (Y)
W <- lalonde[, c("age", "education",
                 "married", "nodegree", "re74", "re75")]
A <- lalonde$treatment # 1 = treated; 0 = control
Y <- lalonde$re78 # 1978 earnings
```

Causal Estimand and Identification

Our target parameter is the Average Treatment Effect (ATE), defined under counterfactual outcomes as $\mathbb{E}[Y^{a=1} - Y^{a=0}|Black = 1]$. This represents the difference in expected 1978 earnings among Black people where $Y^1|Black = 1$ is the counterfactual outcome if one, perhaps contrary to reality, was assigned job training through NSW and $Y^0|Black = 1$ is the counterfactual outcome if one, perhaps contrary to reality, did not get assigned job training (control).

Since treatment was randomly assigned, marginal exchangeability holds by design $Y^a \perp A$. We still require consistency and positivity:

- Consistency: An individual's observed outcome Y when they receive treatment $A = a$ is equal to their counterfactual outcome Y^a.
- Positivity: $0 < p(A = 1 \mid W) < 1$ for observed strata W. Every individual must have a non-zero probability of being assigned or not assigned to the training.

Consider $E[Y^1|Black = 1]$:

- $= E[Y^1|A = 1, Black = 1]]$ (by marginal exchangeability, $Y^a \perp A$)
- $= E[Y|A = 1, Black = 1]]$ (by consistency)

We can replace the 1's with 0's to find $E[Y^0|Black = 1]$. Thus, the causal ATE is identified by the statistical estimand:

$$A\hat{T}E = \mathbb{E}[Y \mid A = 1, Black = 1] - \mathbb{E}[Y \mid A = 0, Black = 1]]$$

Estimating the Average Treatment Effect

We apply three estimators discussed in class to determine estimate the statistical estimand: g-computation (standardization), augmented inverse-propensity of treatment weights (AIPTW), and TMLE with Ensemble SL (van der Laan et al., 2007). AIPTW is doubly-robust, so naturally it should leverage the predicted values from both the outcome model (calculated in the G-computation step) and the propensity score model. Ensemble SL yields flexible estimation of the propensity score and outcome regression, and the doubly-robust TMLE estimator ensures favorable asymptotic properties as long as one of these (propensity score or outcome regression) is correctly specified. Both of these methods are known to be more statistically powerful than G-computation alone.

```
# Fit the outcome model predicting Y from A and W
gcomp_model <- glm(re78 ~ treatment + age + education + married + nodegree + re74 + re75, data = lalonde

# Create two counterfactual datasets
data_A1 <- lalonde
data_A1$treatment <- 1 # All treated

data_A0 <- lalonde
data_A0$treatment <- 0 # All untreated

# Predict the counterfactual outcomes
QAW_1 <- predict(gcomp_model, newdata = data_A1, type = "response") # All treated counterfactual outcom
QAW_0 <- predict(gcomp_model, newdata = data_A0, type = "response") # All untreated counterfactual outc

# Calculate the ATE
ate_gcomp <- mean(QAW_1 - QAW_0) # ATE

# Calculate SE for G-comp using a non-parametric bootstrap (e.g., 200 iterations)
set.seed(123)
B <- 200
n_obs <- nrow(lalonde)
boot_ates <- numeric(B)

for(i in 1:B) {
  # Sample with replacement
```

```r
  boot_indices <- sample(1:n_obs, replace = TRUE)
  boot_data <- lalonde[boot_indices, ]

  # Refit the model on the bootstrap sample
  boot_model <- glm(re78 ~ treatment + age + education
                   + married + nodegree + re74 + re75, data = boot_data, family = gaussian())

  # Create counterfactuals for the bootstrap sample
  boot_A1 <- boot_data; boot_A1$treatment <- 1 # All treated
  boot_A0 <- boot_data; boot_A0$treatment <- 0 # All untreated

  # Predict and calculate the bootstrap RD for get the ATE
  boot_Q1 <- predict(boot_model, newdata = boot_A1, type = "response")
  boot_Q0 <- predict(boot_model, newdata = boot_A0, type = "response")
  boot_ates[i] <- mean(boot_Q1 - boot_Q0)
}

# The SE is the standard deviation of the bootstrap ATEs
se_gcomp <- sd(boot_ates)

# CI for G-computation
lower_gcomp <- ate_gcomp - 1.96 * se_gcomp
upper_gcomp <- ate_gcomp + 1.96 * se_gcomp
ci_gcomp <- paste0("(", round(lower_gcomp, 2), ", ", round(upper_gcomp, 2), ")")

### Estimator 2: Augmented Inverse Probability of Treatment Weighting (AIPTW)

# Fit the propensity score model (g-model)
ps_model <- glm(treatment ~ age + education
                + married + nodegree + re74 + re75,
                data = lalonde, family = binomial())

# Get the predicted probabilities
g1 <- predict(ps_model, type = "response") # Predicted probability of being treated for each individual
g1 <- pmax(0.025, pmin(0.975, g1)) # Trunate for better positivity
g0 <- 1 - g1 # Predicted probability of being untreated for each individual

# Calculate the AIPTW components for the counterfactual means
Y1_aiptw <- (lalonde$treatment / g1) * (lalonde$re78 - QAW_1) + QAW_1
Y0_aiptw <- ((1 - lalonde$treatment) / g0) * (lalonde$re78 - QAW_0) + QAW_0

# Calculate the ATE
ate_aiptw <- mean(Y1_aiptw - Y0_aiptw)

# Calculate SE for AIPTW using the variance of the influence curve
IC_aiptw <- (Y1_aiptw - Y0_aiptw) - ate_aiptw
var_aiptw <- var(IC_aiptw) / n_obs
se_aiptw <- sqrt(var_aiptw)

# CI for AIPTW
lower_aiptw <- ate_aiptw - 1.96 * se_aiptw
upper_aiptw <- ate_aiptw + 1.96 * se_aiptw
ci_aiptw <- paste0("(", round(lower_aiptw, 2), ", ", round(upper_aiptw, 2), ")")
```

Table 1: ATE Estimates on LaLonde Data

Estimator	ATE	CI
G-computation	1371.45	(-281.69, 3024.59)
AIPTW	1417.46	(-408.16, 3243.09)
TMLE	873.43	(-582.24, 2329.11)

```r
### Estimator 3: TMLE with Ensemble SuperLearner (SL)

# Reorder the dataset chronologically since LTMLE will get mad at me: Covariates -> Treatment -> Outcom
lalonde <- lalonde[, c("age", "education", "married",
                          "nodegree", "re74", "re75", "treatment", "re78")]
# Define the SuperLearner library
sl_lib <- c("SL.glm", "SL.mean", "SL.gam", "SL.earth", "SL.xgboost")

# Run TMLE
set.seed(123)
tmle_fit <- suppressWarnings(suppressMessages(ltmle(lalonde,
                  Anodes = "treatment", # Treatment
                  Ynodes = "re78", # Outcome
                  SL.library = sl_lib,
                  abar = list(1, 0))))

# Extract the ATE and CI
tmle_summary <- summary(tmle_fit)
ate_tmle <- tmle_summary$effect.measures$ATE$estimate
tmle_ci_bounds <- tmle_summary$effect.measures$ATE$CI
ci_tmle <- paste0("(", round(tmle_ci_bounds[1], 2), ", ", round(tmle_ci_bounds[2], 2), ")")

### Compile all results

# Compile all results into a single data frame
results_df <- data.frame(
  Estimator = c("G-computation", "AIPTW", "TMLE"),
  ATE = c(ate_gcomp, ate_aiptw, ate_tmle),
  CI = c(ci_gcomp, ci_aiptw, ci_tmle)
)
```

```r
# Output as a table for readability
kable(results_df, format = "latex", digits = 2, caption = "ATE Estimates on LaLonde Data") %>%
  kable_styling(bootstrap_options = c("striped", "hold_position"))
```

G-computation estimates an $\hat{ATE}$ of \$1371.454 (95% CI: -281.69, 3024.59), AIPTW estimates an $\hat{ATE}$ of \$1417.464 (95% CI: -408.16, 3243.09), and TMLE + SL estimates an $\hat{ATE}$ of \$873.432 (95% CI: -582.24, 2329.11).

If unmeasured confounding, marginal exchangeability, consistency and positivity hold, this implies that among Black people, enrollment in the NSW Demonstration increased real earnings in 1978 by more than those who weren't enrolled in the program.

Conclusion

Under consistency, marginal exchangeability, and positivity, this study successfuly identified the causal $A\hat{T}E$ for the effect of NSW Demonstration enrollment on real earnings in 1978 among a small sample of Black individuals. G-computation, AIPTW, and TMLE + SL appear to show increases in earnings with program enrollment for Black individuals, but this finding may or may not be due to chance. The wide confidence interval may be due to low predicted treatment probabilities for certain covariate combinations, or the fact that we do not have access to unmeasured confounders of the effect.

References:

Dehejia, R., and Wahba, S. (1999), "Causal Effects in Nonexperimental Studies: Reevaluating the Evaluation of Training Programs," Journal of the American Statistical Association, 94, 1053-1062.

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Lalonde, R. (1986), "Evaluating the Econometric Evaluations of Training Programs," American Economic Review, 76, 604-620.

Manpower Demonstration Research Corporation, Board of Directors, (1980). "Summary and Findings of the National Supported Work Demonstration". Ballinger Publishing Company.

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